NP comparatives and ellipsis

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Abstract

A new argument is provided for prenominal NP comparatives requiring extraction of the comparative *-er* morpheme out of the DP, accompanied by the late merge of the *than*-clause (cf. Bhatt and Pancheva 2004). We propose that this extraction is driven by the requirements of ellipsis resolution within the *than*-clause. Whenever these requirements are obviated, NP comparatives can instead be interpreted within the DP.

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1 The puzzles

The sentences in (1) are instances of NP comparatives, specifically, of **prenominal NP comparatives.** They differ in the content of the comparative clause, as signaled by the specific auxiliary used. In (1a), the comparison is to the degree of smartness of students that Gal talked to; we call such comparatives '*do*-comparatives' in the following. In (1b), the comparison is to the degree of Charlie's smartness (at a prior time); we call such comparatives '*be*-comparatives'.

(1) a. Gal talked to a smarter student than Charlie did.
    b. Gal talked to a smarter student than Charlie used to be.

The sentence in (1a) requires the *than*-clause to be merged outside the matrix VP, in order to allow for ellipsis resolution: namely, the antecedent for the elided content in the *than*-clause is the VP containing the indefinite in which the *-er* morpheme originates. The *-er* morpheme is generated as the head of the degree phrase (DegP) and undergoes QR to a position outside the VP (and, hence, the DP); the *than*-clause, within which parallel degree abstraction has occurred, is then late merged with it at this higher position; ellipsis resolution is licensed in this configuration since the antecedent-containment has been resolved (cf. Bhatt and Pancheva 2004). The resulting LF is in (2), assuming that DP quantifiers are interpreted *in situ* (at PF, the low copy of *-er* is spelled-out).

(2) [λd Gal talked to [$_{DP}$ a [$_{DegP}$ d-smart student]]] [er $\lambda d'$ than Charlie did ~~talk to a d'-smart student~~]

A standard prediction of this derivation is that any counterpart of (1a) in which extraction out of the DP is blocked should be ungrammatical. This is precisely what we see in the *do*-comparative in (3), where the indefinite determiner has been replaced with a strong universal determiner.

(3) *Gal talked to every smarter student than Charlie did.

This unacceptability of (3) is due to the fact that strong quantifiers like *every*-DPs induce an island for extraction (e.g., Ross 1967, Chomsky 1977). This, in turn, makes it impossible to establish the required configuration to license ellipsis, that is, the LF required for ellipsis resolution incurs an island violation.

So far so good. A puzzle arises regarding the *be*-comparative in (1b). In this sentence, no material external to the DP is required for ellipsis resolution. This observation aligns, for example, with the analysis of NP comparatives on which the AP *smart student* is base-generated within the *than*-clause and raises into the DP, a derivation akin to NP raising in relative clauses (Lechner 2004).

(4) Gal is [$_{DP}$ a [$_{DegP}$ smarter student [λd than Charlie used to be [$_{DegP}$ d [$_{AP}$ ~~smart student~~]]]]]

Given such a derivation, and the apparent independence of the material in the *than*-clause from that in the matrix clause, one would expect the acceptability of *be*-comparatives to show no sensitivity to the nature of the determiner. However, this is not the case, as exemplified in (5).[1]

(5) *Gal talked to every smarter student than Charlie used to be.

These facts are replicated with other weak vs. strong DP hosts of prenominal NP comparatives. For example, there is a contrast between the acceptable negative indefinite in (6a), and the unacceptable definite in (6b) (for the latter observation, see Bresnan 1973, ex. 164).

(6) a. Gal talked to no smarter student than Charlie used to be.
b. *Gal talked to the smarter student than Charlie used to be.

In light of these data, the generalization can be stated as in (7).

(7) *Generalization 1*: Prenominal NP comparatives with a postnominal *than*-clause are acceptable in weak DPs, and unacceptable in strong DPs.

In more theoretical terms, prenominal comparatives with a postnominal *than*-clause uniformly require extraction of the *-er* morpheme out of the DP.

Postnominal NP comparatives exhibit a different behavior than their prenominal counterparts (e.g., Bresnan 1973, Lechner 2004): *be*-comparatives are acceptable in all DPs, including strong DPs, while *do*-comparatives are never acceptable, regardless of the determiner. This is exemplified in (8)-(9).

[1] To our knowledge, this configuration has received little attention in the literature. Keenan 1986, fn. 3 notes that "it is unnatural to say *every taller boy than Bill*." Lechner 2020 cites Beil 1997 and Lerner and Pinkal 1995 for relevant examples. While Beil 1997 includes one such case, marking it as acceptable, Lerner and Pinkal's examples either lack an overt *than*-clause or involve only *do*-comparatives. Bresnan 1973 characterizes the definite variant as marked (pace Beil 1997; see (6b) in the main text), consistent with our findings for *every*. We consulted 4 other native speakers of English, all of whom reported a clear contrast between (1b) and (5).

(8) a. Gal talked to a student smarter than Charlie used to be.

b. Gal talked to every student smarter than Charlie used to be.

(9) a. *Gal talked to a student smarter than Charlie did.

b. *Gal talked to every student smarter than Charlie did.

The acceptability pattern above forces us to draw the following generalization:

(10) *Generalization 2*: Postnominal NP comparatives are uniformly acceptable with *be*-comparatives, and uniformly unacceptable with *do*-comparatives.

Couched in more theoretical terms, this suggests that in postnominal *do*-comparatives, (9), the extraction from the DP required for ellipsis resolution is blocked, whereas in postnominal *be*-comparatives, (8), such extraction is not required. Summarizing, the patterns we need to explain are as in Table 1.

	Prenominal NP comparatives		Postnominal NP comparatives	
	do	*be*	*do*	*be*
Weak determiners	✓	✓	✗	✓
Strong determiners	✗	✗	✗	✓

Table 1: NP comparatives

In Sections 2-3, we derive these generalizations as consequences of an ellipsis-driven process, whereby the *than*-clause must attach outside the DP in certain cases in order to license ellipsis resolution. In Section 4, several correct predictions of the account are discussed. Section 5 concludes the squib by discussing an apparently challenging observation due to Lechner (2004).

2 Prenominal NP comparatives

The analysis of *do*-comparatives provided above, namely that they involve extraction out of the DP, once extended to *be*-comparatives, naturally predicts the two generalizations in (7) and (10). Starting with prenominal NP comparatives, repeated below in (11), let us consider the possible structures for the *than*-clause, ruling them out sequentially. First: The elided predicate in (11) could, in principle, be the adjective alone. In this case, there would be no reason for the comparative clause to attach outside the DP. This is indicated in (12).

(11) Gal talked to a/*every smarter student than Charlie used to be.

(12) *[$_{DP}$ Det λx [[$_{DegP}$ λd x d-smart student] [er $\lambda d'$ than Charlie used to be ~~d'-smart~~]]]

The empirical grounds against the DP having such a structure are provided by Bresnan (1973): the elided constituent must match at least the full degree phrase, including the NP (see, e.g., Abney 1987 on DegP). This is evidenced by the inference that the comparandum must be in the predicate denoted by the DegP: *Gal is a smarter student than Charlie used to be* entails *Charlie used to be a student*. Accordingly, sentence *Gal is a smarter student than her dog used to be* is infelicitous because Gal's dog could not be a student, though it could be smart. In theoretical terms, the antecedent for ellipsis cannot be the adjectival head alone (i.e., *smart*), but must be phrasal (i.e., at least *smart student*).

Second: The elided predicate could, in principle, be the DegP alone, as provided in (13) (cf. Lechner 2004 for a variant). This would, however, mean that the structure intended to satisfy ellipsis licensing is grammatically ill-formed, since the required VP cannot be generated (cf. **Charlie used to be smart student*). While ellipsis is known to obviate certain violations of grammar (see, e.g., island obviations in sluicing, Ross 1969), this does not extend to absence of a licit VP.

(13) *[$_{DP}$ Det λx [[$_{DegP}$ λd x d-smart student] [er $\lambda d'$ than Charlie used to [$_{*VP}$ be ~~d'-smart student~~]]]

Finally, the elided predicate could be a VP containing the indefinite phrase. This, however, means that the *-er* morpheme must move outside of the DP to get an appropriate antecedent for ellipsis, as in (14) (cf., e.g., Warner 1985, Potsdam 1997 on the pronunciation of *be* in VP ellipsis). The elided constituent is well-formed (cf. *Charlie is a smart student*) and has the observed meaning.[2]

(14) [λd Gal talked to [$_{DP}$ a d-smart student]] [er $\lambda d'$ than Charlie used to be ~~a d'-smart student~~]

As (14) is the only viable structure for *be*-comparatives, the unacceptability of the prenominal NP comparative with a strong determiner is explained: in a structure such as (15), movement out of the DP is required, but not possible due to strong DPs being islands. (Even if such movement were possible, the required replacement of the universal determiner with an indefinite in the *than*-clause, as in (15), would violate all common ellipsis licensing conditions.)

(15) *[λd Gal talked to [$_{DP}$ every d-smart student]] [er $\lambda d'$ than Charlie used to be ~~a d'-smart student~~]

[2]The ability to interact with matrix modals further supports *-er* scoping out of the DP in prenominal NP comparatives. Specifically, Stateva/Heim ambiguities replicate: e.g., (ia) permits a reading where Gal's paper must be shorter than Charlie's. This interpretation arises from *less* scoping above the modal, as in (ib) (cf. Stateva 1999, Heim 2000, Bhatt and Pancheva 2004).

(i) a. Gal is allowed to submit a less long paper than Charlie's paper was.
b. [λd [$\Diamond$ Gal submit a d-long paper]] [less $\lambda d'$ Ch.'s paper was ~~a d'-long paper~~]

3 Postnominal NP comparatives

Let us now turn to the generalization in (10). Recall that the pattern with postnominal NP comparatives differs from that observed with prenominal ones: *be*-comparatives are uniformly acceptable, whereas *do*-comparatives are uniformly unacceptable. The crucial difference between pre- and postnominal *be*-comparatives lies in the size of the DegPs in the two cases. The crucial difference in *do*-comparatives lies in whether the *-er* morpheme is generated high in the DP or within a reduced relative clause lower in the DP.

The postnominal *be*-comparative in (8b), repeated below in (16), can be parsed as in (17): the comparative makes up the reduced relative clause modifying the noun, and the DegP consists of the adjective alone. This implies that the ellipsis resolution does not require extraction out of the relative clause.

(16) Gal talked to every student smarter than Charlie used to be.

(17) Gal talked to [$_{DP}$ every [student [$_{RC}$ λx [[$_{DegP}$ λd x d-smart] [er $\lambda d'$ than Charlie used to be ~~d'-smart~~]]]]]

A further consequence of the parse is that, unlike in the prenominal case, the comparative does not entail that Charlie was a student, in line with Bresnan's (1973) observation. This is evidenced by the felicity of (18).

(18) Gal talked to every student smarter than her dog used to be.

In contrast, in the *do*-comparatives in (19), extraction to the matrix level is required as before in order to retrieve the appropriate antecedent for ellipsis resolution, as provided in (20). Since this would now involve extraction out of a relative clause island, the sentences are correctly predicted to be unacceptable, regardless of the quantifier.[3]

(19) *Gal talked to a/every student smarter than Charlie did.

(20) *[λd Gal talked to every/a student [$_{*RC}$ $\lambda x\ x\ d$-smart]] [er $\lambda d'$ than Charlie did ~~talk to a student $\lambda x\ x\ d'$-smart~~]

This concludes the derivation of the two generalizations we set out to address. They emerge from an interplay between what the required antecedent is for ellipsis, and the acceptability of the movement necessary to recover it. The proposal makes several predictions, to which we turn in the next section.

4 Consequences and predictions

Phrasal comparatives. All our examples above involved clausal comparatives, that is, comparatives in which the *than*-phrase is uncontroversially a clause (as witnessed by the presence of an auxiliary). However, the pattern discussed above also extends to phrasal comparatives in English, as shown in (21): the indefinite NP comparative in (21a) allows for both the *do*- and the *be*-reading, while (21b) is ungrammatical.

(21) a. Gal talked to a smarter student than Charlie. [ambiguous]
b. *Gal talked to every smarter student than Charlie.

This is unexpected on the **direct analysis of phrasal comparatives,** which does not posit a clausal structure for the *than*-phrase but rather appropriately modifies the meaning of *-er* so that it combines directly with an individual and a

[3] However, see Sichel (2018) for a recent discussion of counterexamples to relative clause islandhood in Hebrew and other languages. Due to our limited scope, we must postpone a more thorough investigation of the predictions of our account regarding the specific configurations in which obviations of relative clause islands have been observed across languages.

degree predicate to yield the meaning observed above for clausal comparatives (see, e.g., Heim 1985, Kennedy 1997; see Bhatt and Takahashi 2011 for a recent discussion). On this account, the LF structure that would be assigned to the sentence in (21b) would simply be of the form in (22).

(22) Gal talked to [every [$_{DegP}$ [smart student] [er than Charlie]]]

As with the raising analysis in Lechner (2004) mentioned above, this structure is well-formed – therefore, (21b) should be acceptable. In other words, the direct analysis of phrasal comparatives makes an incorrect prediction with respect to their behavior in prenominal cases. In contrast, on the **ellipsis analysis of phrasal comparatives,** on which *than*-phrases in (21) are underlyingly clausal, the pattern in (21) is correctly predicted, fully parallel to our derivation above. This provides a new argument for a clausal analysis of English phrasal comparatives, on the one hand. On the other hand, it gives rise to a prediction: if there are languages where phrasal comparatives are in fact interpreted directly, the counterpart of (21b) is predicted to be acceptable, all else being equal.

One example of a language that appears to have phrasal comparatives requiring a direct analysis is Slovenian. In Slovenian, there are two ways to realize the first argument of the *-er* morpheme. The first involves a *kot*-prefixed clause – that is, the phrase that *kot* heads may contain clausal material (e.g., it may have discontinuous remnants); this makes it akin to English *than*-phrases. The second way involves the preposition *od* 'from'. In this case, the phrase cannot contain any material other than a genitive-marked DP (see, e.g., Pancheva 2006, Berezovskaya and Hohaus 2015, on related constructions in other Slavic languages and arguments for their direct analysis). As predicted by our account, this phrasal comparative is acceptable with prenominal NP comparatives, regardless of whether the host DP is an indefinite as in (23a)[4] or a strong quantifier, as in (23b).

(23) a. Pametnejši študent od Petra je govoril z Gal.
smarter student from Peter$_{gen}$ aux talked with Gal
"A smarter student than Peter talked to Gal."

b. Vsak pametnejši študent od Petra je govoril z Gal.
every smarter student from Peter$_{gen}$ aux talked with Gal
"*Every smarter student than Peter talked to Gal."

The clausal counterparts of these sentences, in contrast, are acceptable only if the host DP is an indefinite, that is, they are unacceptable if the DP is a universal quantifier, as illustrated in (24). This is as expected.

(24) a. Pametnejši študent kot (je) Peter je govoril z Gal.
smarter student than (aux) Peter$_{nom}$ aux talked with Gal
"A smarter student than Peter talked to Gal."

b. *Vsak pametnejši študent kot (je) Peter je govoril z Gal.
every smarter student than (aux) Peter$_{nom}$ aux talked with Gal
"*Every smarter student than Peter talked to Gal."

Bare comparatives. Another prediction of our proposal is that if ellipsis requirements do not force the *-er* morpheme to take scope outside the DP, prenominal NP comparatives should be acceptable, regardless of the determiner. An obvious configuration in which this is the case are 'bare' comparatives, that is, comparatives without an overt *than*-phrase. In these examples, the first argument of the *-er* morpheme is contextually supplied and no silent structure of its complement need be assumed. The *-er* morpheme and its argument can thus be interpreted within the DP. It has been observed that these are indeed acceptable, as shown in (25) (e.g., Lerner and Pinkal 1995, Beil 1997).

[4] As Slovenian is an articleless language, the indefiniteness is inferred from the interpretation.

(25) Gal talked to every smarter student.

5 Outlook

We provided an argument for an analysis of prenominal NP comparatives that relies on a late merger of *than*-clauses outside the DP hosting the comparative at surface form (cf. Bhatt and Pancheva 2004). Specifically, this allowed us to account for the two generalizations presented in Section 1 as well as to draw correct predictions for proper phrasal comparatives in languages that have them, and for *than*-less NP comparatives.

Going forward, one avenue for future investigation pertains crucially to the assumption of late and high attachment of the *than*-clause and its various repercussions involving predicted c-command relations. One such repercussion, noted by Lechner (2004), appears to militate against the proposal. Specifically, Lechner observes that a comparative sentence like (26) is acceptable under the bound-variable interpretation of the pronoun in the *than*-clause.

(26) Gal offered a better description of each law$_i$ than its$_i$ author could.

Under the proposal described above, the analysis of the sentence requires exceptional binding of *its* by the universal quantifier, analogous to the patterns discussed under the heading of inverse linking (e.g., May 1977). The LF yielding this interpretation is provided in (27).

(27) [each law λx [[λd Gal offered a d-good description of x] [er $\lambda d'$ than its$_x$ author could ~~offer a d'-good description of x~~]]]

The structure involves binding by *each law* of a variable that is not c-commanded by *each law* in its base position. Nonetheless, this configuration does not obviously constitute a violation of weak crossover, strictly speaking, given that *each law* does not bind anything to the left of its A-position (cf. Postal 1971, Chomsky 1976, Jacobson 1977). Accordingly, the structure should be well-formed, all else equal, and it yields the observed interpretation.

Lechner (2004), however, challenges the generality of such a derivation on the basis of a pair of observations: (i) inverse linking across a negated indefinite does not seem to be possible, as illustrated in (28), yet (ii) it would be required for the reading of the corresponding NP comparative sentence, provided in (29). In light of this, he concludes that the attachment of the *than*-clause cannot be higher than the NP in the general case, as required by our analysis.

(28) *Gal didn't expect any description of each law$_i$ to characterize it$_i$ properly.

(29) Gal couldn't come up with any better description of each law$_i$ than its$_i$ author.

We concur that the asymmetry in acceptability between (28) and (29) on the indicated binding is real and challenging. We contend, however, that observations (i-ii) exemplified in (28)-(29) do not pose a challenge for our proposal per se, but rather for any theory of NP comparatives, including that of Lechner (2004),[5] as well as for a proper formulation of the conditions under which inverse linking is possible. In particular, closer scrutiny of (29) reveals that the sentence receives precisely the interpretation corresponding to the inverse linking reading. More to the point, not only is the universal quantifier in (29) not interpreted within the indefinite, as is intuitively clear, it is not even interpreted within the scope of negation. To appreciate this, consider the scenario in (30).

[5] In particular, note that (29) involves a *do*-comparative interpretation, requiring QR of the *-er* outside the DP on all existing analyses, and a subsequent QR of *each law* to obtain the indicated binding at LF.

(30) Scenario: Gal came up with a better description of Law1 than Law1's author, and she came up with better description of Law2 than Law2's author. But she did not came up with a better description of Law3 than Law3's author.

The sentence in (29) is judged false in this scenario, even though it should be judged as true if the universal quantifier were interpreted in the scope of negation. We conclude that although much work remains, at least some apparent challenges to the proposed analysis are just that – apparent.

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